



Effects of Lévy noise on transport of active Brownian particles

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HIGHLIGHTS

- The transport properties of active Brownian particles driven by Lévy noise are studied.
- Performance is strongly sensitive to Lévy index α and noise strength of Lévy noise.
- The optimal value of Lévy index α for which the mean positive velocity shows minimum may be important for other investigations.

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ABSTRACT

The transport properties of active Brownian particles driven by Lévy noise are investigated. It is found the Lévy index α plays a significant role in the transport of the active Brownian particles, where both the mean velocity and the mean escape time show the non-monotonic dependence on the Lévy index. The effective diffusion increases as the Lévy index α increases.

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1. Introduction

During the past two decades, the constructive role of noise has been widely observed in physical, chemical, as well as in biological systems. Examples manifesting constructive role of noises include stochastic resonance [1–3], spatiotemporal order out of noise [4], and noise-induced transport [5,6], etc. The transport properties of Brownian particles affected by noise have been investigated extensively due to the possible applications in various fields, such as molecular motors [7–11], nanoscale friction [12,13], surface smoothening [14], and so on. In the simplest situations, it is assumed that the noise acting on a test particle is white and Gaussian. In far from equilibrium situations, the stochastic process describing interactions of the test particle with the environment can still be of the white type but its values can be distributed according to some heavy tailed distribution. Lévy flight, which denotes a class of non-Gaussian stochastic processes, can be observed in many different situations ranging from the detection of neurons faint or subthreshold signals [15], self-diffusion in micelle systems [16–18], super-diffusion in a magnetic field [19], genetic regulatory [20], and many others [21–23]. Due to heavy-tail characteristics and infinite divisibility, the dynamical systems subjected to the Lévy noise exhibit quite unusual and fascinating properties. Among these studies, one important issue is to explore the non-equilibrium effect of the Lévy noise in the Lévy ratchet. J.D. Bao pointed out that the presence of α -stable Lévy type noise affects the escape process of the particles and the escape rate shows nonmonotonic dependence on the Lévy noise [24]. Dybiec and co-workers [25,26] studied the minimal setup for a Lévy ratchet and found that due to the nonthermal character of the Lévy noise, the net current can be obtained even in the absence of whatever additional time-dependent forces. Rosa and Beims [27] studied the optimal transport and its

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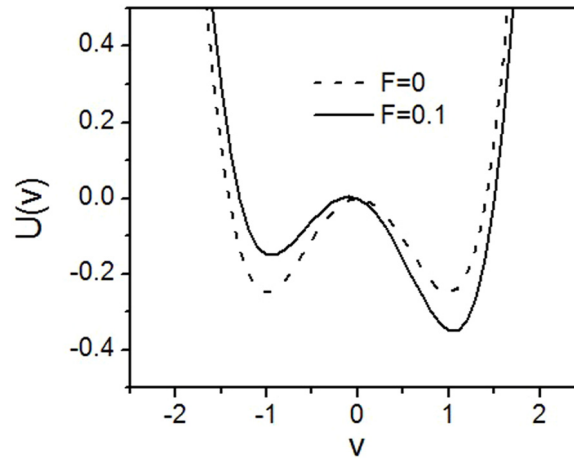


Fig. 1. The velocity potential of the active Brownian particles for the symmetric case ($F = 0$) and the asymmetric case ($F = 0.1$)

relation to superdiffusive transport and Lévy walks for Brownian particles in ratchet potential in the presence of modulated environment and external oscillating forces. B.Q. Ai studied the transport of Brownian particles in the presence of ac-driving forces and Lévy noises, and multiple current reversals were observed [28].

Recently, the transport properties of active Brownian particles (ABP) have attracted much attention [29,30] since it can explain the origin of the self-propelled motion, and the active(self-propelled) motion is one of the obvious signatures of many living systems and can be observed on different levels, ranging from the motion of motor proteins within cells to the motility of cells and bacteria [31–33]. Our system is closely related to the one investigated in Ref. [34], where the mean drift and effective diffusion of such a system are discussed when a simple Gaussian noise is applied. However, Lévy noise has advantages over standard Gaussian noise in many models though it increases mathematical complexity. Therefore, the effects of the Lévy noise on the transport properties of the active Brownian particles need to be further investigated.

In this paper, the transport properties of the active Brownian particle driven by Lévy stable noise are investigated. In Section 2, the dynamical model of the active Brownian particles with Lévy noise is presented. In Section 3, the effects of Lévy noise on the transport of active Brownian particles are discussed. A discussion concludes the paper.

2. Dynamical model and theoretical analysis

The Langevin equation of the active Brownian particle driven by Lévy stable noise can be written as follows [35–37]

$$\begin{aligned} \dot{x} &= v \\ \dot{v} &= -\frac{dU(v)}{dv} + \xi_\alpha(t) \end{aligned} \quad (1)$$

Here all parameters and variables in Eq. (1) are dimensionless. $x(t)$ is the position of the particle at time t . The biased velocity potential $U(v) = v^4/4 - v^2/2 - Fv$ is plotted in Fig. 1 where the asymmetry is introduced by a bias F . It is shown that the potential is symmetric for $F = 0$, and asymmetric for $F = 0.1$. Since the influence of the bias F on the transport of active Brownian particles has already been reported in Ref. [34], in the following discussion F is fixed to be 0.1 for a non-zero velocity. The Lévy noise $\xi_\alpha(t)$ is a white, symmetric α -stable noise possessing Lévy stable distribution $\lambda(x) \sim \frac{\sigma^\alpha}{|x|^{1+\alpha}}$ with $0 < \alpha \leq 2$, where σ is a scaling factor of physical dimension length $[\sigma] = \text{cm}$. [38] In the Fourier domain, the characteristic function $\lambda(k)$ of $\lambda(x)$ can be written as [39,40]

$$\lambda(k) = F\{\lambda(x)\} = \int_{-\infty}^{\infty} e^{ikx} \lambda(x) dx = \exp(-D|k|^\alpha). \quad (2)$$

Here $D = \sigma^\alpha$ denotes the strength of noise [41].

The simulation strategy of Eq. (1) can be given as [42]

$$\begin{aligned} x_{n+1} - x_n &= v_n \\ v_{n+1} - v_n &= -\frac{dU(v_n)}{dv} \Delta t + (D\Delta t)^{1/\alpha} \xi_\alpha(n). \end{aligned} \quad (3)$$

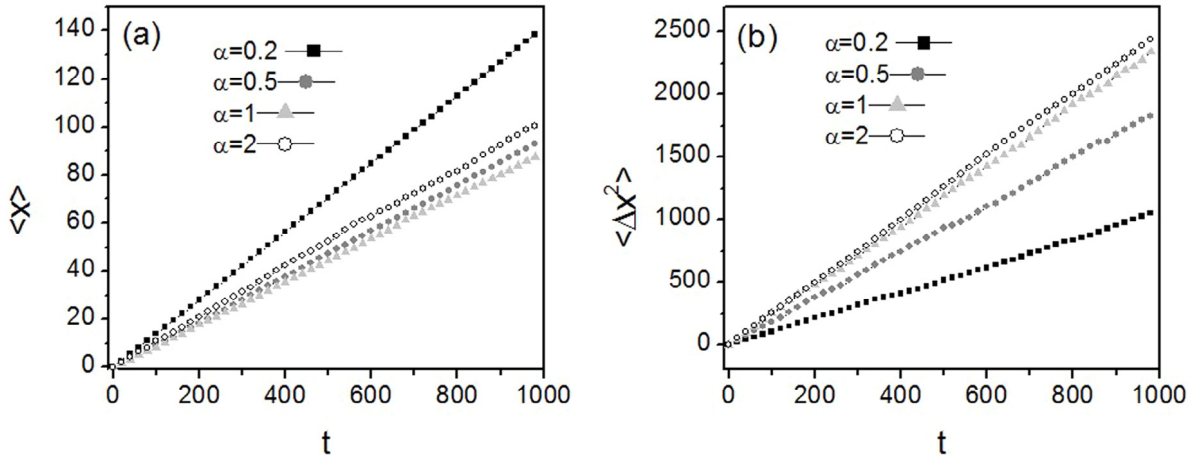


Fig. 2. The first moment $\langle x(t) \rangle$ and variance of the displacement $\langle \Delta x^2(t) \rangle$ of ABP as a function of time t for different values of Lévy index α with Lévy noise strength $D = 1$.

where $\xi_\alpha(n)$ is a set of random variables corresponding to the characteristic function (2) and can be generated as [42]

$$\xi_\alpha = \frac{\sin(\alpha\gamma)}{(\cos\gamma)^{1/\alpha}} \left(\frac{\cos[1-\alpha]\gamma}{W} \right)^{(1-\alpha)/\alpha}, \quad (4)$$

where γ is a random value uniformly distributed in the interval $(-\pi/2, \pi/2)$, and W is an independent random variable which is exponentially distributed. α is the Lévy index with $0 < \alpha \leq 2$. For $\alpha = 2$, the Lévy stable noise becomes a standard Gaussian white noise.

The fractional Fokker–Planck equation can be written as [43,44]

$$\frac{\partial P(v, t)}{\partial t} = -\frac{\partial}{\partial v} [f(v) P(v, t)] + D \frac{\partial^\alpha}{\partial v^\alpha} P(v, t), \quad (5)$$

where $f(v) = -\frac{dU(v)}{dv} = v - v^3 + F$. For $\alpha = 2$, the Lévy stable noise becomes a standard Gaussian white noise, that is:

$$\frac{\partial P(v, t)}{\partial t} = -\frac{\partial}{\partial v} [f(v) P(v, t)] + D \frac{\partial^2}{\partial v^2} P(v, t), \quad (6)$$

Then the velocity distribution $P_{st}(v)$ can be derived as

$$P_{st}(v) = \frac{e^{-U(v)/D}}{\int_{-\infty}^{\infty} dv e^{-U(v)/D}}. \quad (7)$$

The mean velocity and the diffusion coefficient can be calculated by [34]

$$\langle v \rangle = \int_{-\infty}^{\infty} v P_{st}(v) dv, \quad (8)$$

$$D_{eff} = \frac{\int_{-\infty}^{\infty} dx e^{-U(x)/D} \left[\int_{-\infty}^x (y - \langle v \rangle) e^{-U(y)/D} dy \right]^2}{D \int_{-\infty}^{\infty} e^{-U(v)/D} dv}. \quad (9)$$

3. Properties of active Brownian particles

The numerical simulations of Eqs. (3) and (4) are performed by using the simulation strategy proposed by Ref. [42] with a time step of $\Delta t = 10^{-2}$. The data are saved over 3000 different trajectories.

3.1. Displacement and velocity distribution

The first moment $\langle x(t) \rangle$ and the mean-square displacement $\langle \Delta x^2(t) \rangle$ of ABP as a function of time t are plotted in Fig. 2 for different values of Lévy index α , where the mean square displacement of the particle is $\langle \Delta x^2(t) \rangle = \langle x^2(t) \rangle - \langle x(t) \rangle^2$. It is found that both the first moment $\langle x(t) \rangle$ and the mean-square displacement $\langle \Delta x^2(t) \rangle$ exhibit a linear function of time with different slopes as Lévy index α increases. Therefore, the mean velocity can be determined by $\langle v \rangle = \langle x(t) \rangle / t$, where $\langle v \rangle$ represents the drift velocity. The effective diffusion can be represented by $D_{eff} = \lim_{t \rightarrow \infty} \langle \Delta x^2(t) \rangle / (2t)$ [38].

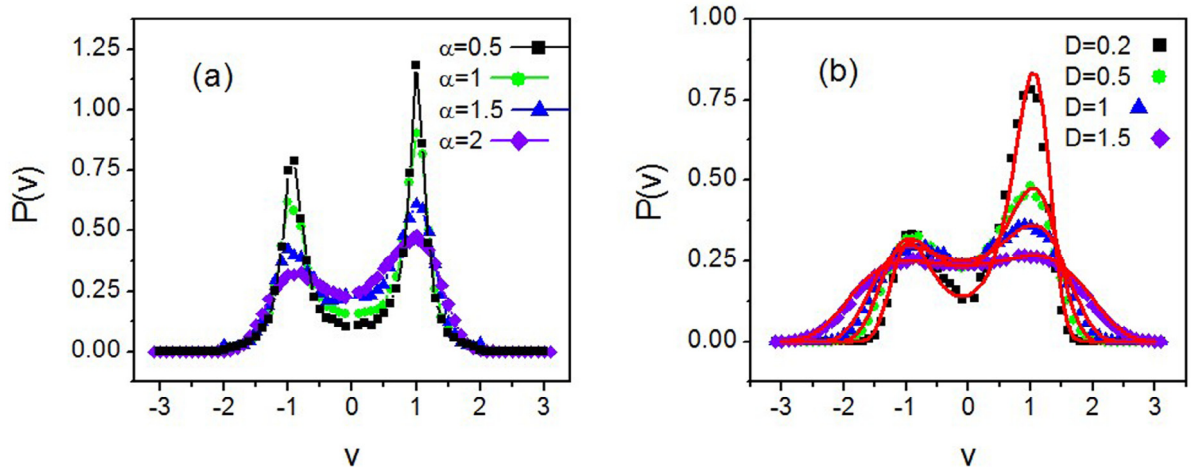


Fig. 3. (a) The probability distribution $P_{st}(v)$ for different values of Lévy index α with $D = 0.5$. (b) The probability distribution $P_{st}(v)$ for different values of Lévy noise strength D with the Gaussian case $\alpha = 2$. Symbols correspond to numerical simulations from a direct integration of Eq. (3) and Red lines are the theoretical results of Eq. (7). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

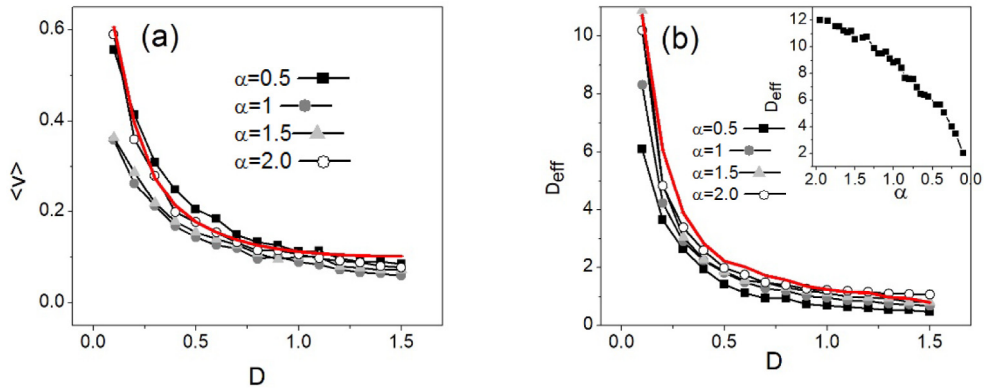


Fig. 4. (a) Dependence of the mean velocity $\langle v \rangle$ on the Lévy noise strength D for different values of Lévy index α . (b) Dependence of the effective diffusion D_{eff} on the Lévy noise strength D for different values of Lévy index α . Symbols correspond to numerical simulations from a direct integration of Eq. (3) and Red lines are the theoretical results of Eq. (9). Inset: Dependence of the effective diffusion D_{eff} on the Lévy index α . (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

The numerical simulations of the velocity distribution $P_{st}(v)$ are plotted in Fig. 3(a) for different Lévy index α . From Fig. 3(a), it is shown that when the Lévy index α increases, the two peaks at $v \approx \pm 1$ decrease gradually, while the valley at $v \approx 0$ increases. To check the validity of the numerical approach, the theoretical results as well as the numerical simulations of the velocity distribution $P_{st}(v)$ for the case of the standard Gaussian white noise $\alpha = 2$ are plotted in Fig. 3(b) for different noise strength D . Symbols correspond to numerical simulations from a direct integration of Eq. (3) and lines are the theoretical results of Eq. (7). It is clear that theoretical results agree well with the numerical simulations. It seems that when the noise strength D increases, both the left peak at $v \approx -1$ and the right peak $v \approx 1$ decrease and finally there are two peaks at $v = -1$ and $v = 1$ with almost the same height.

3.2. Mean velocity and effective diffusion

The mean velocity $\langle v \rangle$ and the effective diffusion D_{eff} as a function of the noise strength D are plotted in Fig. 4 for different Lévy index α . For $\alpha = 2$, the theoretical predictions of the mean velocity $\langle v \rangle$ from Eq. (8) and the diffusion coefficient D_{eff} from Eq. (9) are plotted as red lines. It is found that the mean velocity $\langle v \rangle$ shows positive value and both the mean velocity $\langle v \rangle$ and the effective diffusion D_{eff} decreases monotonically as the noise strength increases. According to the fluctuation theorem, higher value of the noise is correlated with a higher diffusion, which has been reported in many non-active Brownian motion systems [45,46]. However, the diffusion coefficient here shows a surprisingly decreasing dependence on the noise strength. This phenomenon seems commonly exist in active Brownian models [34,47]. There exists a critical value of the bias F ($|F_c| \approx 0.2$, in Fig. 5(a) and $|F_c| \approx 0.4$ in Fig. 5(b)), below which the diffusion coefficient D_{eff} decreases as the Gaussian

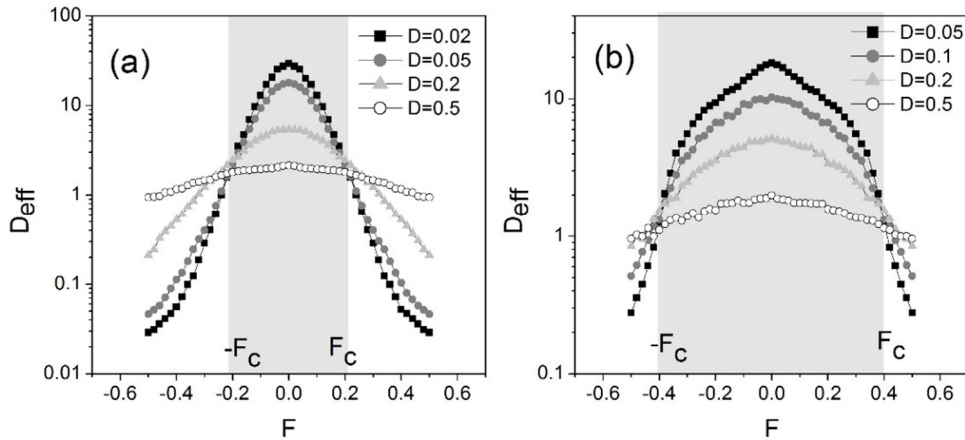


Fig. 5. (a) Dependence of the effective diffusion D_{eff} on the bias F for different Lévy noise strength D with the Gaussian case $\alpha = 2$. (b) Dependence of the effective diffusion D_{eff} on the bias F for different Lévy noise strength D with the Gaussian case $\alpha = 1$.

white noise increases, as shown in the inset of Fig. 5. The Lévy noise does not affect this trend, but the diffusion coefficient decreases as comparing with the Gaussian case ($\alpha = 2$), shown in the inset of Fig. 4(b).

The mean velocity $\langle v \rangle$ and the effective diffusion D_{eff} as a function of the Lévy index α are plotted in Fig. 6 for different noise strength D . From Fig. 6(a), it is found that the mean velocity $\langle v \rangle$ decreases initially as α increases, attains minimum at $\alpha = 1$, and then increases as α increases further. In Fig. 6(b), it is shown that the effective diffusion D_{eff} increases monotonically as the Lévy index α increases.

The mean escape time T_{esc} of the particles as a function of the Lévy index α for different Lévy noise strength D is plotted in Fig. 7, where the particles are initially located in the potential well $v(t=0) = -1$, the time of particles takes to overcome a threshold at $v = 0$ is measured by the mean escape time. A larger mean escape time means a poorer ability for the particles to escape from $v(t=0) = -1$ to the threshold $v = 0$. Non-monotonic dependence of T_{esc} on the Lévy index α can be observed. All the curves of T_{esc} have an apparent maximum at an optimal value of $\alpha \approx 1$. It seems that the Lévy index $\alpha = 1$ is a critical value for which the mean velocity v shows minimum of Fig. 6(a) and the mean escape time T_{esc} shows maximum. This can be understood as follows: as the α increases, the mean escape time T_{esc} increases at first, which makes it increasingly difficult for particles to escape from $v = -1$ to $v = 0$. As more and more particles accumulate at $v(t=0) = -1$ and less likely to escape, the positive mean velocity v decreases when α increases from 0.1 to 1. Then the mean escape time T_{esc} reaches the maximum at $\alpha = 1$, which is related to the corresponding minimum in the mean velocity v , shown in the inset of Fig. 6. The mean escape time T_{esc} decreases for further increasing α , which means it is a bit easier for particles to escape from $v = -1$ to $v = 0$, and consequently an increase in the mean velocity v .

4. Discussion

In conclusion, the effects of Lévy noise on the transport properties of the active Brownian particles are investigated. It is found both the mean velocity and effective diffusion exhibit a monotonically decreasing dependence on the noise strength. It should be mentioned that the monotonically decreasing dependence of the effective diffusion on the noise strength goes against the fluctuation dissipation theorem. There exists a critical value of the bias F , below which the diffusion coefficient decreases as the noise increases. There exists an optimal value of the Lévy index $\alpha = 1$ for which the mean positive velocity is minimum. The effective diffusion increases as the Lévy index α increases. Moreover, it is found that the mean escape time shows a maximal value at $\alpha = 1$. Such mechanism manifests that the Lévy index α plays a significant role in the transport of the active Brownian particles, the non-monotonic dependence of the mean velocity v on α is related to the non-monotonic dependence of T_{esc} on α . Our investigations may be helpful in understanding the self-propelled motion of biological process and the optimal value of the Lévy index α for which the mean positive velocity shows minimum may be important for engineers, and imply a possible strategy for the control of transport.

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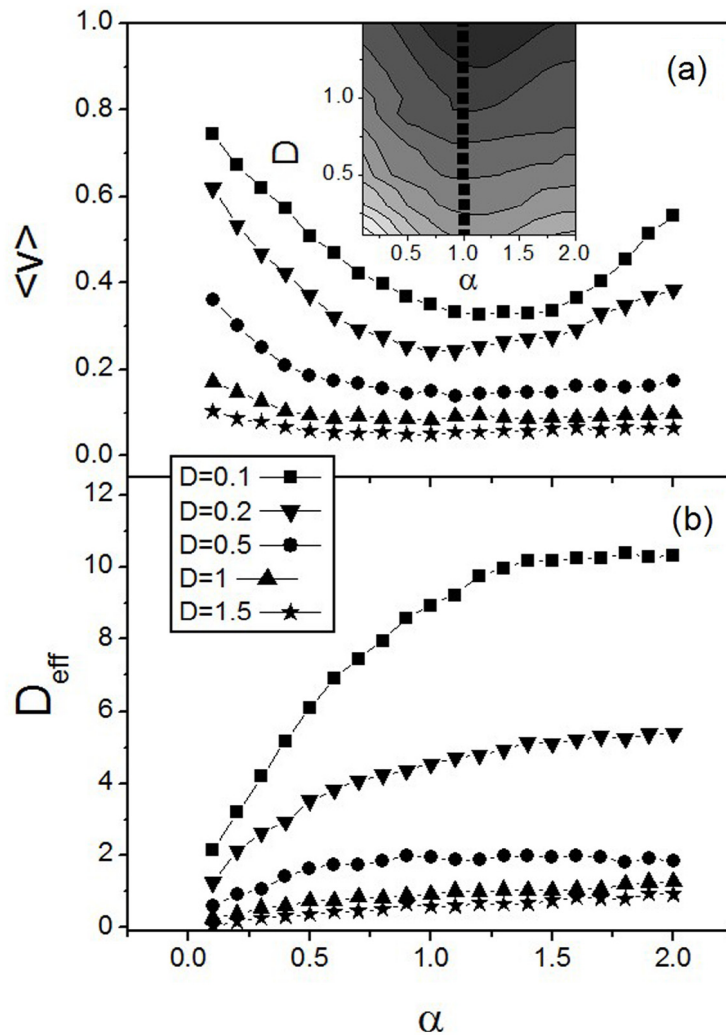


Fig. 6. (a) Dependence of the mean velocity $\langle v \rangle$ on the Lévy index α for different Lévy noise strength D . Inset: The contour plot of $\langle v \rangle$ as a function of the Lévy index α and Lévy noise strength D . (b) Dependence of the effective diffusion D_{eff} on the Lévy index α for different Lévy noise strength D .

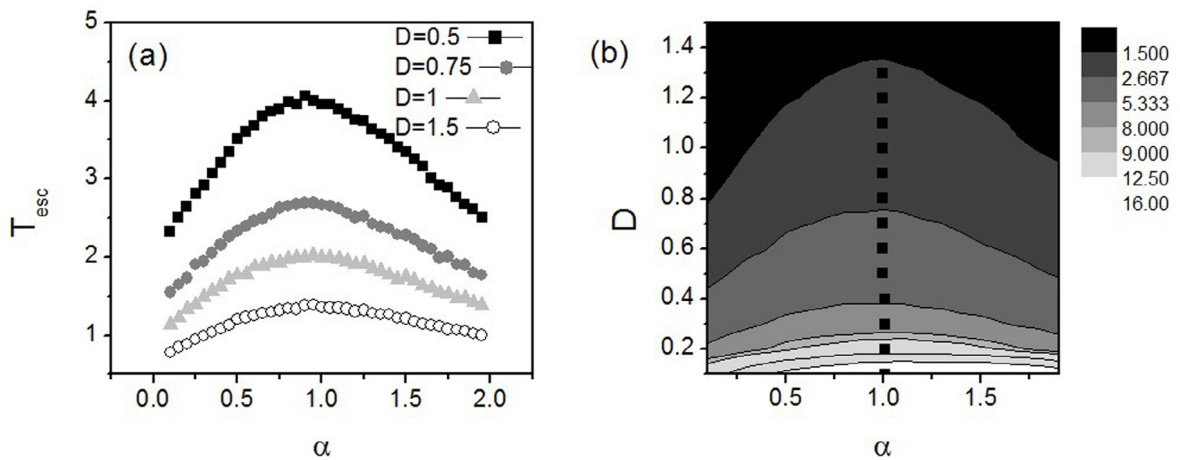


Fig. 7. (a) The mean escape time T_{esc} as a function of the Lévy index α for different Lévy noise strength D . (b) The contour plot of T_{esc} as a function of the Lévy index α and Lévy noise strength D .

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